Tshana Tumelo Konaite

Machine Learning Engineer

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Professional Summary

AI Engineer with an MSc in Computational and Applied Mathematics specializing in machine learning systems, statistical modelling, and LLM-powered AI solutions. Experienced in designing and deploying production-ready AI systems, including RAG pipelines and multi-agent architectures using Python and cloud-native infrastructure.

Education: MSc Computational and Applied Mathematics — University of the Witwatersrand

Tech Stack: Python, SQL | Pandas, NumPy, Scikit-learn | LangChain, OpenAI SDK | FastAPI | AWS, Docker | Git, CI/CD

AWS Certifications: Generative AI Developer – Professional | Machine Learning Engineer – Associate | Solutions Architect – Associate

Professional Experience

ZenoStudy — Machine Learning Engineer (Dec 2025 – Present)

- Built scalable AI systems powered by LLMs to automate learning workflows and knowledge generation.
- Evaluated foundation models using LLM-as-a-Judge and retrieval metrics (MRR, Recall@k, Precision@k), alongside latency and cost benchmarking to optimize model selection.
- Developed production-grade Python APIs using FastAPI, enabling reliable and scalable AI-driven services.
- Designed RAG pipelines and multi-model orchestration systems, improving content generation speed and enabling scalable deployment of AI applications.

Huawei Technologies — Data Analyst (Dec 2019 – Nov 2025)

- Developed an automated meeting intelligence pipeline using speech transcription and LLM-based summarization, reducing report turnaround time from 2 days to 30 minutes.
- Engineered a retrieval-based AI assistant enabling new team members to query internal datasets conversationally, improving knowledge access and onboarding efficiency.
- Automated geospatial data transformations, significantly accelerating planning cycles and reporting turnaround times.
- Built a machine learning model for telecom infrastructure optimization, identifying cellular towers requiring upgrades using signal strength, latency, and throughput data.

Projects

- Built an AWS-powered multi-agent pricing system that evaluates marketplace listings using ensemble models to estimate fair value and identify underpriced products.
- Designed a predictive analytics model for customer churn, analyzing customer behavior patterns to identify high-risk users and support targeted retention strategies.
- Built a healthcare AI application that converts doctor consultation notes into structured medical summaries, actionable next steps, and patient-ready email communications using real-time AI generation.